

Comparative Analysis of Machine Learning, Deep Learning, And Transformer Models for Social Media Sentiment Analysis

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Abstract — The exponential growth in the amount of data on social media, which is composed of large amounts of unstructured and human text containing opinions, feelings and trends, has turned the analysis of this data into an essential matter in different areas such as health, finance and marketing. However, the nature of social media texts that are expressed in an informal and uncontrolled manner poses a significant obstacle to the correct classification. In this study, a comparison study has been made between machine learning and deep learning models for sentiment analysis on the Sentiment140 dataset. For this purpose, the Naive Bayes, Support Vector Machine (SVM), Long Short-Term Memory (LSTM) and RoBERTa models were implemented and the results were compared. The performance of the models was evaluated using accuracy, precision, recall, F1-score, ROC and PR curves. The results show that transformer-based and deep learning models have better performance than the traditional models for analyzing contextual and noisy data, and also indicate the computational costs associated with them.

Keywords — *Sentiment Analysis, Social Media, Machine Learning, Deep Learning, LSTM, RoBERTa, NLP, Sentiment140*

I. INTRODUCTION

Social media platforms are rapidly turning into the biggest source of user-generated content, where users share their thoughts, feelings and opinions constantly throughout the day. Social media platforms such as Twitter have become an important medium for sharing views on social issues in the form of short text messages. Therefore, there is a need to analyze these tweets on an ongoing basis. Sentiment analysis, also known as opinion mining, is a subfield of Natural Language Processing, which analyzes text to identify whether opinions expressed in it are positive or negative. It is applicable to a variety of applications such as customer reviews for an e-commerce website, health monitoring, stock exchange and commodity trading.[1], [2]

Sentiment analysis is a relatively well-studied area of research. However, analysing the sentiment of data from sources such as social media is a significantly more challenging task. The primary reason for this is that social media contains a large amount of unstructured free-form text, which often contains a large amount of slang, acronyms, emojis, as well as unnecessary or poorly spelt words. A further challenge to the task is the short length of tweets. This means that the text rarely contains any information that could be classed as excess, which can be used to determine the intent of the classification. A further challenge that arises in the classification of the intent of the text is the large amount of sarcasm and irony contained in such datasets. Words can be used to express a meaning that is different to their literal definition. This problem is especially prevalent in datasets sourced from social media [4], [5].

In recent years, a large amount of research has been done on Sentiment Classification using machine learning techniques, where the widely used algorithms for this purpose are Naive Bayes and Support Vector Machine (SVM) [6], [11]. They use statistical features such as Term Frequency-Inverse Document Frequency (TF-IDF) and look at words as individual features. While these approaches work well with data with a certain level of structure

(such as reviews), they do not perform as well with unstructured and highly noisy datasets, such as social media, where context is key to sentiment classification.

This basic approach, however, has two main shortcomings: it heavily depends on the selected features and is not very effective with long sentences. In order to achieve better results, deep learning models are used. Using the knowledge of the sequence of features and their context, they yield much better results. LSTM (Long Short-Term Memory) is an example of such a model, which not only recognises the relationships between words in a sequence, but also is aware of the sentence structure and semantics. More advanced models like the transformer-based BERT (Bidirectional Embedding Representations from a Transformer) and RoBERTa are able to achieve very good results[5], [8]. They apply the concept of the attention mechanism to understand both ways in the text. Many researches has shown that deep learning models, especially in the field of NLP has been leading in areas such as sentiment analysis, when dealing with large and complex datasets.

As impressive as modern machine learning developments have become, a review of the literature indicates a lack of studies comparing multiple approaches comprehensively. Nearly all of them study only a single paradigm in isolation: Either the “classic” ML approaches or state-of-the-art Deep Learning approaches. Some works concentrate on the advantages of modern approaches like the transformer, others highlight the ease-of-use and effectiveness of basic approaches, but only a few works appear to compare every single approach with respect to a single dataset and applied preprocessing as well as metrics. The latter is often unclear to determine, as several conclusions in the literature imply that the modern approaches always outperform the classic models. It is therefore hard to tell by how much the former is superior to the latter in real-world applications.[5], [13]

The existence of linguistic ambiguities, noisy communications and heterogeneous terminology in social media forums has become a serious impediment to understanding public sentiment. Hence this study endeavours to propose a sentiment analysis method with the ability to resist the impact of noises in social media and compare the performance of various machine learning and deep learning approaches. In this work, we trained machine learning approaches such as Naive Bayes and SVM, deep learning approaches including LSTM, and transformer approach including RoBERTa by using the Sentiment140 dataset and evaluated their performances using the metrics of accuracy, precision, recall, F1-score and graphical evaluation measures such as ROC curve and precision-recall curve. Through the comparative study conducted under uniform experimental design in this research, the aims are to develop an optimal method for the sentiment analysis and understand the underlying communication behaviours.

II. RELATED WORK

Nowadays, huge amounts of social media content have pushed sentiment analysis into sharper focus across many studies. At first, researchers mostly leaned on classic machine learning approaches - think Naive Bayes, SVM, or Logistic Regression. To turn raw text into usable inputs, those systems relied on methods like bag-of-words or TF-IDF. Even though these tools work quickly and keep

things straightforward, they tend to fall short by ignoring how words relate to one another. Because each word gets handled alone, deeper meaning in sentences slips through the cracks [6], [7].

A new wave of deep learning arrived to work around earlier flaws. What stood out? Long Short-Term Memory networks - LSTM for short - became top performers in understanding human language tasks; those examples before used them[8]. These systems differ because they hold onto past word details much longer than older versions, making it easier to grasp how sentences fit together [8]. Research keeps showing a pattern: when built on varied, massive data pools, LSTM-driven approaches tend to boost precision nearly every time.

One reason newer language systems stand out is how they process words using structures called transformers. Instead of moving step by step through sentences, these designs look at surrounding text all at once. Attention layers let each word pull meaning from what comes before and after it. Performance jumps were seen when models began reading longer stretches during learning. Tasks like judging if a review feels positive or negative show clear gains with these approaches. Better results came not from changing the model shape but adjusting how long and how often training runs. Among them, RoBERTa gained an edge by refining those settings further.[5], [8]

Looking at feelings has helped in many areas too. Patient comments get examined in health settings, while emotions in illness discussions are tracked closely [2]. From tweets to headlines, mood clues have guided guesses about market shifts and how investors react [3]. Real world uses prove this tool brings value beyond theory.

Still, some problems remain. When social media data isn't cleaned well before use, the model works poorly - this kind of data tends to be messy and casual [4]. Many current studies instead look at just one model at a time, missing broader comparisons. Because evaluations differ so much, spotting the top performer in matching situations becomes difficult [11], [12].

Heavy computing demands mark a key issue with deep learning and transformers. Though these models boost precision, their reliance on vast processing power and long training stretches limits broad use[13]. Even if newer methods outperform older ones, studies stress comparing various machine learning techniques side by side - same data, same rules for measuring results.

III. METHODOLOGY

This research is not only pertinent to the sentiment mining aspect on social media but also relevant to the comparison of effectiveness between Machine Learning and Deep Learning methodologies. Our methodology consists of several steps: from selecting the appropriate dataset, data preprocessing, to extracting text representation features and developing the appropriate models to handle the inherent noise within the social media environment.

A. Data

Sentiment140 is one of the most well known benchmarks for sentiment analysis in human language processing. For this study a subset of roughly 1.6 million tweets was used, and the polarity labelling of the text corresponds to the notion of the study about the relation between expression and sentiment, where 0 denotes negative and 4 denotes positive sentiment.[1],[2]

We used samples from the Sentiment140 data set to evaluate the trade-off between efficiency, performance and balance for this work. We used a sample of 200,000 tweets for this study. The samples were processed text and sentiment labels, and came in a single data file. Note that there is no neutral class included in the data set. The data was structured into training data and test data based on an 80:20 ratio. Each section of the samples were designated by a change in the sentiment classes and shifted from a skewed distribution of the sentiment classes to a relatively balanced distribution.

B. Data Preprocessing

Social media data often comes in the form of noisy text consisting of plain symbols such as hashtags and usernames. Preprocessing the data before analysis is required in order to clean and normalise it, in order to improve the model's accuracy.

We change all the text in the data to lowercase so that we can perform comparison easily. URL, @ (user mention), # (hash tag), special characters, punctuation, etc. are removed because these won't be of any use in determining the sentiment of the statement. I also removed the stop words which makes the code more efficient because by removing these it reduces word redundancy thereby making it easier to focus on words that are actually relevant.

Furthermore, applying the principles of data science such as handling of extra spaces and repeated characters, the normalised form of given text is produced. These steps are very important for machine learning and deep learning models because noisy input can affect their accuracy drastically.[13]

C. Text Representation and Tokenization

We pre-process the textual data, this is followed by numerical conversion so that different models can process the data. Tokenization is used in a variety of models. Let's go over a few of them.

- The machine learning models used in the research employ TF-IDF vectorization in order to transform the text into a numerical matrix that captures the frequency of words across all the texts in the database as well as the importance of those words. While this method captures everything relating to word frequency, it does not capture the word order or relationships between words.[6], [11]
- The code utilizes the Keras tokenizer from deep learning models where each word is assigned a unique integer and text sequences are converted to a sequence of integers. Padding is necessary due to the varying length of tweets and it ensures that all samples have a uniform sequence length
- In RoBERTa, we use the pretrained RoBERTa tokenizer. The tokenizer in RoBERTa uses Byte Pair Encoding (BPE) to split the text into subword tokens. BPE tokens improve the handling of rare and unknown words while preserving the context for the meaning of words..[5], [8]

D. Feature Extraction

In sentiment analysis, a crucial step in preparing data for training a model is the feature extraction. This is the step where the textual data is transformed into numerical values to be used as input by a machine learning algorithm.

In our approach, we used TF-IDF features of preprocessed text as input vectors to machine learning models. This reflects the importance of words within the given dataset.

In deep learning models, the tokenised input sequences are embedded into dense vector space through an embedding layer. This significantly improves the performance of a model by capturing complex semantic relationships between individual tokens.

Transformer-based models typically generate their contextual embeddings within the model itself by means of an attention mechanism. This allows to incorporate both semantic and contextual information in order to create more powerful models.

E. Model design

We present results for the four defined models in terms of sentiment analysis in different setups.

For the sentiment analysis task in the machine learning project we employed two models. The first one was a Naïve Bayes model, a probabilistic machine learning model which is based on the assumption of independent features. This model is computationally efficient and was therefore chosen for this task. The second model is a Support Vector Machine (SVM), a widely used supervised learning model that aims to find the optimal decision boundary in order to separate two classes. We chose SVM for this task because it is known for performing well in high-dimensional data.

To tackle the temporal challenges of text classification in the deep learning era, we employed LSTM. In our LSTM-based model, it captures temporal dynamics in text with awareness of sequential context, word co-occurrence, and semantics. To further improve performance, we also built a transformer-based RoBERTa model, which leverages attention to capture both forward and backward contexts for a much deeper text semantics.

F. Methodology Pipeline

From the previous steps we have selected a dataset, cleaned the text and have started the process of turning the clean text into numerical form using the relevant tokeniser for the relevant model. This is followed by the step of feature extraction. We then proceed to feed the input through our models. Once we have done this we can then train and evaluate the models.

The overall workflow of the proposed system for sentiment analysis is visualized in Fig. 1. First, the input text is preprocessed. Then, features are extracted, the model is trained and evaluated.

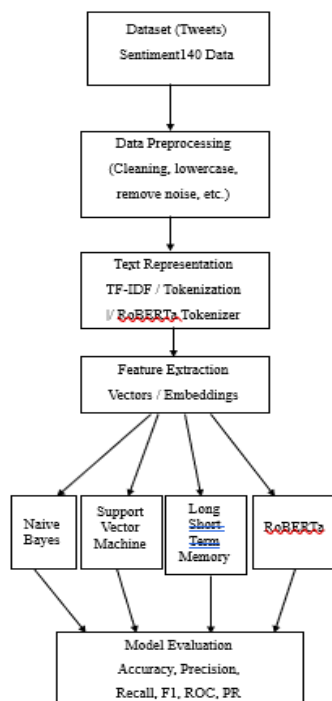


Fig. 1: Proposed sentiment analysis pipeline for model comparison.

IV. IMPLEMENTATION

The proposed Sentiment Analysis System was written in Python and utilizes both machine learning and deep learning

frameworks. The code is implemented in the Google Colab environment, which uses GPU to speed up the deep learning architecture.[6], [5]

We have utilized 200k tweets out of Sentiment140. After that, we did preprocess to remove noise and converted the entire data into lowercase. Then it is divided into Training set and Testing Set with a ratio of 80% - 20%.

For machine learning models, TF-IDF Vectorizer is used to convert the Text Data into Numerical Features. Naive Bayes was chosen as the baseline model and SVM for better Classification.

LSTM Deep Learning Architecture was constructed using TensorFlow and Keras. Text Data was tokenized into Sequences and Padded to make sure all Input has Same Length. The embedding layer is used to generate a dense representation. LSTM Layer is used to learn Sequence Dependencies. And the Sigmoid Activated Output Layer is used for Binary Classification.

The transformer model is implemented using RoBERTa. Pretrained Tokenizer from Hugging Face Transformers Library is used to encode the Text. Model is Fine-Tuned for Sentiment Classification.

Training and Evaluation Conditions for All Models are Consistent. Accuracy, Precision, Recall and F1-Score are used to Evaluate Performance. ROC Curve and Precision – Recall Curve are also used to analyze the detailed performance.

V. RESULT ANALYSIS

The performance of the models was evaluated using accuracy, precision, recall, and F1-score, along with ROC and precision–recall curves. All models were tested on the same dataset for a fair comparison.[5], [13]

Model	Accuracy	Precision	Recall	F1-score
Naïve Bayes	0.7762	0.7766	0.7742	0.7754
SVM	0.7987	0.7900	0.8126	0.8012
LSTM	0.9489	0.9493	0.9483	0.9488
RoBERTa	0.9552	0.9533	0.9572	0.9552

Table 1. Performance comparison of sentiment analysis models.

From Table.1, we can clearly see that RoBERTa achieves the highest performance, followed by LSTM, while SVM and Naïve Bayes show lower performance.

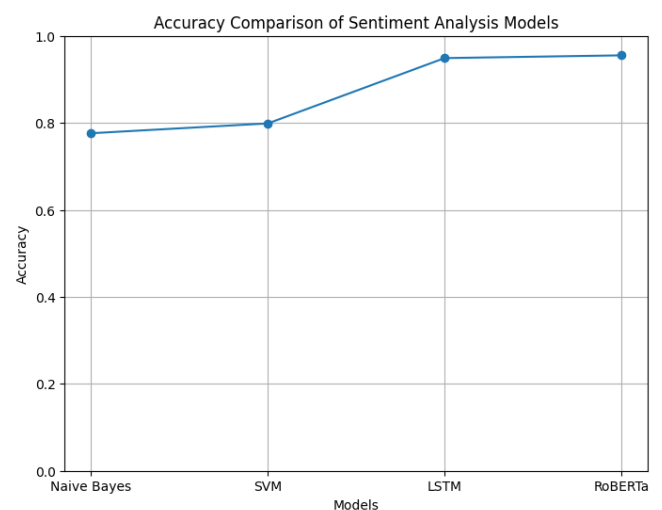


Fig. 2. Accuracy comparison of sentiment analysis models.

The accuracy graph clearly shows that RoBERTa and LSTM outperform traditional models.

From Fig.3, the ROC curves also support these findings. LSTM and RoBERTa have curves closer to the top-left corner, which indicates better performance. Their AUC values are higher compared to Naïve Bayes and SVM.

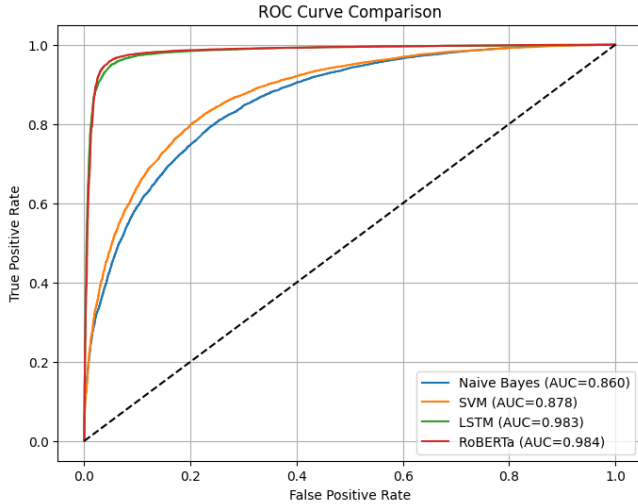


Fig. 3. ROC curve comparison of models.

Similarly, from Fig. 4, the precision–recall curves show that LSTM and RoBERTa maintain high precision even when recall increases. In contrast, traditional models show a drop in performance. This means advanced models handle noisy and difficult data better.

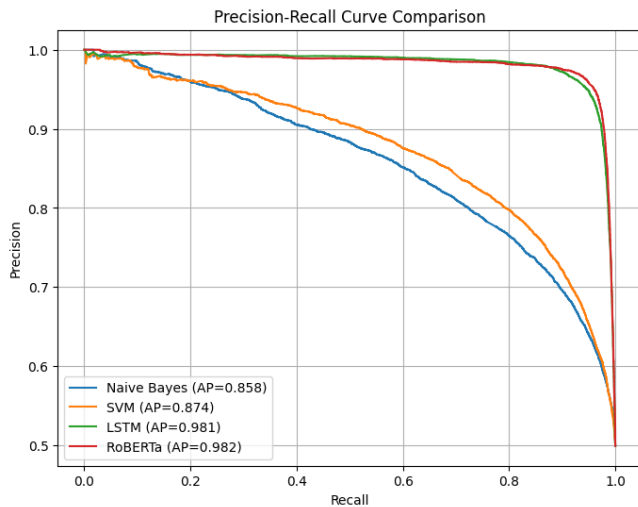


Fig. 4. Precision–recall curve comparison of models.

From the above analysis, we can say that the RoBERTa performs the best, followed by the LSTM.

VI. CONCLUSION

In this research, a comparative study of various models of machine learning, deep learning, and transformers was carried out for sentiment analysis using social media data. Various models, such as Naïve Bayes, SVM, LSTM, and RoBERTa, were implemented and tested based on their performance using various metrics.

From the results, it is evident that traditional models offer a moderate level of performance, while deep learning models, such as LSTM, offer a high level of accuracy by incorporating contextual information. Among all models, the best performance was offered by the RoBERTa model, which proves that transformer models are effective in processing and understanding complex and unclean data from social media platforms.

From this research, it is evident that advanced models are more suitable for performing sentiment analysis tasks, especially with unstructured data, while traditional models are useful for their simplicity and low computational costs.

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